

# Asymptotics of $n$ -universal lattices over number fields

Quadratic Lattices in Prague, Charles University

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# Preliminaries

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Throughout this talk, every  $\mathcal{O}_K$ -lattice is *classical* and *totally positive definite* (unless otherwise specified).

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- $I_4 \perp \gamma I_4$ , with  $\gamma = D + \sqrt{D^2 - 1}$ , is universal over  $\mathbb{Z}[\sqrt{D^2 - 1}]$ , for any squarefree  $D^2 - 1$ . (Kim 2000)

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Here the subscript  $*$  records what conditions  $L$  must satisfy:

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| $m_{\text{nc}}(\mathcal{O}_K)$      | not necessarily classical (i.e. no condition)      |
| $m_{\text{cl}}(\mathcal{O}_K)$      | classical                                          |
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For each totally real  $K$ :  $3 \leq m_{\text{nc}}(\mathcal{O}_K) \leq m_{\text{cl}}(\mathcal{O}_K) \leq m_{\text{cl,free}}(\mathcal{O}_K) \leq m_{\text{diag}}(\mathcal{O}_K)$ .



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## Conjecture (Kitaoka 1990s)

*There are only finitely many totally real fields  $K$  such that  $m_{\text{cl}}(\mathcal{O}_K) = 3$ .*

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- (Krásenský–Tinková–Zemková 2020) For all totally real biquadratic  $K$ ,  $m_{\text{cl,free}}(\mathcal{O}_K) \geq 4$ .

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- (Kala–Krásenský–Park–Yatsyna–Žmija 2025) If  $K = \mathbb{Q}(\sqrt{D})$  is real quadratic,  $D$  squarefree, and  $m_{\text{nc}}(\mathcal{O}_K) \leq 3$ , then  $D \in \{2, 3, 5, 6, 7, 10, 13, 17, 21, 33, 41, 65, 77\}$ .



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# Minimal ranks: the story so far

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# $n$ -universal $\mathcal{O}_K$ -lattices

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## $n$ -universal quadratic lattices

A (classical, totally positive definite)  $\mathcal{O}_K$ -lattice  $L$  is  **$n$ -universal** if it represents every  $\mathcal{O}_K$ -lattice of rank  $n$  (i.e. if for every totally positive rank  $n$   $\mathcal{O}_K$ -lattice  $M$ , there is an  $\mathcal{O}_K$ -linear  $\sigma : M \rightarrow L$  with  $Q_M = Q_L \circ \sigma$ ).

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We write  $U_{\mathcal{O}_K}(n)$  for the minimal rank of an  $n$ -universal  $\mathcal{O}_K$ -lattice.



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Theorem (J. Yoon 2026+)

$$42 \leq U_{\mathbb{Z}}(11) \leq 45.$$

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We want to show that  $U_{\mathbb{Z}}(n)$  grows very quickly as  $n \rightarrow \infty$ .



# Asymptotics for $U_{\mathcal{O}_K}(n)$

## Theorem (Park–V–Yatsyna–Yoon 2025)

*Let  $K$  be a totally real degree  $d$  number field of discriminant  $\Delta_K$ . Then, with the exception of a finite set of at most 559 real quadratic fields  $K$ , we have*

$$\log U_{\mathcal{O}_K}(n) = \frac{d}{4}n^2 \log n + n^2 \left( \frac{2 \log \Delta_K - 2d \log(2\pi) - 3d}{8} \right) + \mathcal{O}(n \log n)$$

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- In fact the  $\leq$  holds for all totally real fields  $K$ , and the  $\geq$  holds for infinitely many  $n$ .

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$$\text{mass}_K(L) \longrightarrow \text{cl}_K(L) \longrightarrow b_{\mathcal{O}_K}(m) \longrightarrow a_{\mathcal{O}_K}(m) \longrightarrow U_{\mathcal{O}_K}(n)$$

# Step 1: Siegel's mass formula

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- For any  $\mathcal{O}_K$ -lattice  $L$ , Siegel defined the following genus-invariant constant:

$$\text{mass}_K(L) := \sum_{\Lambda \in \text{gen}(L)} \frac{1}{\#\text{Aut}(\Lambda)}$$

where  $\text{gen}(L)$  denotes the (finitely many) isometry classes of  $\mathcal{O}_K$ -lattices on  $L \otimes K$  that are everywhere locally isometric to  $L$ .

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- Since  $L$  is positive definite,  $\text{Aut}(\Lambda)$  is finite and contains  $\pm 1$ , thus we have the trivial bounds:

$$2 \text{mass}_K(L) \leq \text{cl}_K(L) \leq \text{mass}_K(L) \cdot \max_{\Lambda \in \text{gen}(L)} \#\text{Aut}(\Lambda).$$

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## Theorem (Siegel's mass formula (1935, 1937))

*Let  $L$  be a positive definite unimodular  $\mathcal{O}_K$ -lattice of rank  $n \geq 2$ . Then*

$$\text{mass}_K(L) = \Delta_K^{n(n-1)/4} \cdot \frac{1}{\alpha_\infty(L)} \cdot \prod_{\mathfrak{p}} \frac{1}{\alpha_{\mathfrak{p}}(L)},$$

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Here,  $\alpha_\infty(L)$  and  $\alpha_{\mathfrak{p}}(L)$  are the local densities:

$$\alpha_\infty(L) := \left( \frac{\pi^{n(n+1)/4}}{2\Gamma(\frac{1}{2})\Gamma(\frac{2}{2}) \cdots \Gamma(\frac{n}{2})} \right)^d,$$
$$\alpha_{\mathfrak{p}}(L) := \lim_{\nu \rightarrow \infty} \frac{\#\{X \in \text{Mat}_n(\mathcal{O}_K/\mathfrak{p}^\nu) : X^T G_{\mathfrak{p}} X \equiv G_{\mathfrak{p}} \pmod{\mathfrak{p}^\nu}\}}{2N(\mathfrak{p})^{\nu n(n-1)/2}},$$

where  $G_{\mathfrak{p}}$  is a Gram matrix of the localisation  $L_{\mathfrak{p}}$ .

# The mass of $I_n$

## Theorem (Körner 1973)

Let  $K$  be a totally real field and let  $n \geq 3$  be an odd integer. For each dyadic prime  $\mathfrak{p}$  in  $K$ , let  $e_{\mathfrak{p}}$  and  $f_{\mathfrak{p}}$  be the ramification index and residue degree of  $\mathfrak{p}$  respectively. Let  $\xi(n, \mathfrak{p})$  be defined as

$$\frac{\xi(n, \mathfrak{p})}{N(\mathfrak{p})^{(1-n)\lfloor e_{\mathfrak{p}}/2 \rfloor - e_{\mathfrak{p}}}} = \begin{cases} 1 & \text{if } 2 \mid e_{\mathfrak{p}}, \\ \frac{1}{2} \left( 1 + \frac{(-1)^{f_{\mathfrak{p}} \lfloor (n+1)/4 \rfloor}}{N(\mathfrak{p})^{\lfloor (n-1)/2 \rfloor}} \right) & \text{if } 2 \nmid e_{\mathfrak{p}}. \end{cases}$$

Then

$$\text{mass}_K(I_n) = \left( \frac{2\Gamma(\frac{1}{2})\Gamma(\frac{2}{2}) \cdots \Gamma(\frac{n}{2})}{\pi^{n(n+1)/4}} \right)^{[K:\mathbb{Q}]} \Delta_K^{n(n-1)/4} \prod_{\mathfrak{p}|2} \xi(n, \mathfrak{p}) \prod_{i=1}^{\lfloor (n-1)/2 \rfloor} \zeta_K(2i),$$

where  $\Gamma(s)$  is the usual Gamma function,  $\zeta_K(s)$  is the Dedekind zeta function of  $K$ .



# Asymptotics for $\text{mass}_K(I_n)$

## Theorem

Let  $K$  be a totally real degree  $d$  field, and for each dyadic prime  $\mathfrak{p}$  in  $K$ , let  $e_{\mathfrak{p}}$  denote the ramification degree of  $\mathfrak{p}$ . Then

$$\text{mass}_K(I_n) \sim C_K \left( \frac{n}{2\pi e \sqrt{e}} \right)^{dn^2/4} \left( \frac{8\pi e}{n} \right)^{dn/4} \left( \frac{1}{n} \right)^{d/24} \Delta_K^{n(n-1)/4} \left( \prod_{\mathfrak{p}|2} \frac{1}{N(\mathfrak{p})^{\lfloor e_{\mathfrak{p}}/2 \rfloor}} \right)^n$$

as  $n \rightarrow \infty$ . Here  $C_K$  is the explicit constant

$$C_K = 2^{-5d/4} e^{d/24} \exp \left( \frac{1}{12} - \zeta'(-1) \right)^{-d/2} \cdot \left( \prod_{\mathfrak{p}|2} \frac{N(\mathfrak{p})^{\lfloor e_{\mathfrak{p}}/2 \rfloor - e_{\mathfrak{p}}}}{2^{\delta_{2 \nmid e_{\mathfrak{p}}}}} \right) \cdot \prod_{i=1}^{\infty} \zeta_K(2i)$$

where  $\delta_{2 \nmid e_{\mathfrak{p}}}$  is 1 if  $e_{\mathfrak{p}}$  is odd, and 0 otherwise.

## Step 2: Bounding automorphism groups

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**Theorem (Schur 1905, odd  $\ell$ )**

*Let  $G$  be a finite  $\ell$ -subgroup of  $\text{GL}_n(\mathbb{C})$  such that  $\text{Tr}(g) \in K$  for all  $g \in G$ . Then:*

$$v_\ell(\#G) \leq m_\ell \left\lfloor \frac{n}{t_\ell} \right\rfloor + \left\lfloor \frac{n}{\ell t_\ell} \right\rfloor + \left\lfloor \frac{n}{\ell^2 t_\ell} \right\rfloor + \dots$$

*where  $t_\ell := [K(\zeta_\ell) : K]$  and  $m_\ell$  is the maximal integer  $a$  such that  $K(\zeta_\ell)$  contains  $\zeta_{\ell^a}$ .*

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### Corollary

Let  $L$  be a positive definite  $\mathcal{O}_K$ -lattice. Then  $\log \# \text{Aut}(L) \leq n \log n + O_K(n)$ .

## Step 3: Counting rank $n$ unimodular $\mathcal{O}_K$ -lattices

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where  $\mathcal{G}_n$  is the set of genera of rank  $n$  unimodular  $\mathcal{O}_K$ -lattices. Applying the known bounds for the  $\text{mass}_K(L)$ ,  $\#\text{Aut } L$ , and the upper bound  $\#\mathcal{G}_n \leq (4\sqrt{5})^d$ , we obtain the following theorem:

### Theorem

*For any totally real field  $K$ ,*

$$\log b_{\mathcal{O}_K}(n) = \frac{d}{4}n^2 \log n + n^2 \left( \frac{2 \log \Delta_K - 2d \log(2\pi) - 3d}{8} \right) + \mathcal{O}(n \log n)$$

*as  $n \rightarrow \infty$ .*

## Step 4: Counting indecomp. unimodular lattices

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Let  $a_{\mathcal{O}_K}(n)$  be the number of isometry classes of *indecomposable* rank  $n$  unimodular  $\mathcal{O}_K$ -lattices.

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$$\prod_{m=1}^{\infty} (1 - X^m)^{-a_{\mathcal{O}_K}(m)} = 1 + \sum_{m=1}^{\infty} b_{\mathcal{O}_K}(m) X^m.$$

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### Theorem (Wright 1968)

Let  $\{T_n\}_{n \geq 1}$  be the Euler transform  $\{t_n\}_{n \geq 1}$ . If  $\sum_{s=1}^{n-1} H_s H_{n-s} = o(H_n)$  as  $n \rightarrow \infty$ , where  $\{H_n\}_{n \geq 1}$  is either  $\{T_n\}$  or  $\{t_n\}$ , then  $T_n \sim t_n$  as  $n \rightarrow \infty$ .

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- We verified that Wright's theorem holds for  $H_n = b_{\mathcal{O}_K}(n)$  over all totally real fields  $K$ , with the exceptional of 559 real quadratic fields  $K$ .

## Step 5: Bounds for $U_{\mathcal{O}_K}(n)$

### Lemma

Let  $K$  be totally real field. Let  $\overline{\mathcal{I}}(m)$  denote all indecomposable unimodular  $\mathcal{O}_K$ -lattices of rank at most  $m$ . Then

$$\sum_{L \in \overline{\mathcal{I}}(n)} \left\lfloor \frac{n}{\text{rank } L} \right\rfloor (\text{rank } L) \leq U_{\mathcal{O}_K}(n) \leq \sum_{L \in \overline{\mathcal{I}}(n+3)} \left\lfloor \frac{n+3}{\text{rank } L} \right\rfloor \cdot (\text{rank } L).$$

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*Sketch of lower bound:*

- Use uniqueness of orthogonal decomposition to show that any  $n$ -universal  $\mathcal{O}_K$ -lattice  $M$  represents

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*Sketch of upper bound:*

- Prove that  $M$  represents every unimodular lattice on  $I_{n+3} \otimes K$ , where  $M$  is

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By the previous lemma, we have

$$\log a_{\mathcal{O}_K}(n) \leq \log U_{\mathcal{O}_K}(n) \leq \log b_{\mathcal{O}_K}(n+3) + 2 \log(n+3).$$

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Applying the upper bound obtained for  $b_{\mathcal{O}_K}(n)$  in Step 3, and the lower bound obtained for  $a_{\mathcal{O}_K}(n)$  in Step 4, this proves the claimed asymptotic for  $U_{\mathcal{O}_K}(n)$ , so we are done!  $\square$

## Pfeuffer's results.

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## Theorem (Pfeuffer 1979)

*Let  $C > 0$ . Then there are only finitely many totally real fields  $K$  such that there exists an  $\mathcal{O}_K$ -lattice  $L$  of rank  $\geq 3$  with  $\text{cl}_K(L) < C$ . Each such field  $K$  admits only a finite number of lattices  $L$  with bounded scale.*

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## Theorem (Pfeuffer 1979)

*Let  $C > 0$  and fix a positive integer  $d$ . Then there are only finitely many totally real fields  $K$  of degree  $d$  such that there exists an  $\mathcal{O}_K$ -lattice  $L$  of rank 2 with  $\text{cl}_K(L) < C$ . Only finitely many of all these fields (of arbitrary degree) can be reached by a tower of relatively normal extensions from  $\mathbb{Q}$ . There are finitely many such fields  $K$  in all, provided either the generalized Riemann hypothesis or the Artin conjecture (on the analyticity of Artin  $L$ -functions) is true.*

# Extending Pfeuffer to minimal ranks

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## Theorem (Park–V–Yatsyna–Yoon 2025)

*Let  $C > 0$  and fix some positive integer  $n \geq 3$ . Then there are only finitely many totally real fields  $K$  such that  $U_{\mathcal{O}_K}(n) < C$ . In particular, all such fields  $K$  satisfying  $U_{\mathcal{O}_K}(n) < C$  are contained within an effectively computable finite set of fields.*

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If  $n \geq 3$  and  $U_{\mathcal{O}_K}(n) < C$ , then the degree  $d$  and discriminant  $\Delta_K$  of such fields  $K$  satisfy the following bounds:

$$d \leq \frac{4n \log(C) + 8 \log(n!) + 254n(n-1) - 4 \log(2)}{\log((2 \cdot 60.1^{n(n-1)/4} \prod_{i=1}^n \Gamma(i/2)) / (2^{(n+5)/2} \cdot \pi^{n(n+1)/4} \cdot (2\zeta(n/2))^{\delta_{2|n}}))},$$
$$\Delta_K \leq \left( \frac{C^n (n!)^2 \pi^{dn(n+1)/4} 2^{d(n+5)/2} (2\zeta(n/2))^{d\delta_{2|n}}}{2 \cdot (2 \prod_{i=1}^n \Gamma(i/2))^d} \right)^{4/(n(n-1))},$$

where  $\delta_{2|n} = 1$  if  $n$  is even, and 0 otherwise.

## Case $n = 2$

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### Theorem (Park–V–Yatsyna–Yoon 2025)

*Let  $C > 0$ , and fix some positive integer  $d \geq 1$ . Then there are only finitely many degree  $d$  totally real fields  $K$  such that  $U_{\mathcal{O}_K}(2) < C$ . In particular, if  $d \geq 3$ , then all such fields  $K$  satisfying  $U_{\mathcal{O}_K}(2) < C$  are contained within an effectively computable finite set of fields.*



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If  $d \geq 3$  and  $U_{\mathcal{O}_K}(2) < C$ , the discriminant  $\Delta_K$  of such fields  $K$  satisfies

$$\Delta_K \leq \max \left( \exp((12(d-1))^2), \left( \frac{2C^2 \cdot (\pi \cdot 2^{5/2})^d \cdot d \cdot (2d)!}{0.000181 \cdot ((d-1)/e)^{d-1}} \right)^{\frac{12d}{5d-12}} \right).$$

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Moreover, there are only finitely many such fields  $K$  in total (of arbitrary degree) which have a normal tower from  $\mathbb{Q}$ . There are only finitely many such fields  $K$  in total, if either the generalised Riemann hypothesis or the Artin conjecture (on the analyticity of Artin  $L$ -functions) is true.

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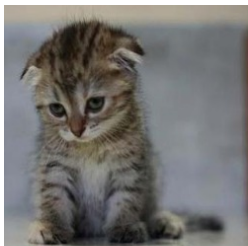
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Being able to  
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# Criterion sets

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A **criterion set** over  $\mathcal{O}_K$  is a set  $\mathcal{S} \subset \mathcal{O}_K^+$  such that an  $\mathcal{O}_K$ -lattice  $L$  is universal if and only if  $L$  represents every element of  $\mathcal{S}$ .

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- A classical  $\mathbb{Z}$ -lattice is universal if and only if it represents every element of  $\mathcal{S} = \{1, 2, 3, 5, 6, 7, 10, 14, 15\}$ . (Conway-Schneeberger 1993; Bhargava 2000)

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- A classical  $\mathbb{Z}[\frac{1+\sqrt{5}}{2}]$ -lattice is universal if and only if it represents every element of  $\mathcal{S} = \{1, 2, \frac{5+\sqrt{5}}{2}, \frac{7+\sqrt{5}}{2}, \frac{7-\sqrt{5}}{2}, \frac{10+2\sqrt{5}}{2}, \frac{15+3\sqrt{5}}{2}\}$ . (Lee 2008)

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- (Kim–Kim–Oh 1999) A  $\mathbb{Z}$ -lattice is 2-universal if and only if it represents:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

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- (Oh 1999) A  $\mathbb{Z}$ -lattice is 8-universal if and only if it represents both  $I_8$  and  $E_8$  (the unique positive-definite even unimodular lattice of rank 8).

# Criterion set theorem

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Theorem (Park–V–Yatsyna–Yoon 2025)

*Let  $C > 0$  and let  $n \geq 3$ . Then there are only finitely many totally real fields  $K$  which admit an  $n$ -universal criterion set  $\mathcal{S}_K(n)$  such that  $\#\mathcal{S}_K(n) < C$ .*

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## Theorem (Park–V–Yatsyna–Yoon 2025)

*Furthermore, for any fixed  $d \geq 1$  there are only finitely many degree  $d$  totally real fields  $K$  which admit a 2-universal criterion set  $\mathcal{S}_K(2)$  such that  $\#\mathcal{S}_K(2) < C$ .*

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- In particular, if  $\#\mathcal{S}_K(n) < C$  then the degree  $d$  and discriminant  $\Delta_K$  of  $K$  satisfy the same upper bounds as given in the previous slide.

## Theorem (Park–V–Yatsyna–Yoon 2025)

*Furthermore, for any fixed  $d \geq 1$  there are only finitely many degree  $d$  totally real fields  $K$  which admit a 2-universal criterion set  $\mathcal{S}_K(2)$  such that  $\#\mathcal{S}_K(2) < C$ .*

- In particular, if  $d \geq 3$  and  $\#\mathcal{S}_K(2) < C$ , then the discriminant  $\Delta_K$  of  $K$  satisfies the same upper bound as given in the previous slide.



## Further problems

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- Can one prove that  $\log U_{\mathcal{O}_K}(n) \sim \frac{d}{4} n^2 \log n$  as  $n \rightarrow \infty$  for the remaining 559 exceptional real quadratic fields  $K$ ?
- For any fixed totally real field  $K$ , can we derive an explicit asymptotic for  $U_{\mathcal{O}_K}(n)$  itself (and not just  $\log U_{\mathcal{O}_K}(n)$ ) as  $n \rightarrow \infty$ ?
- Can we derive bounds for  $U_{\mathcal{O}_K}(1)$ ?
- Can  $U_{\mathcal{O}_K}(n+1) - U_{\mathcal{O}_K}(n)$  be arbitrarily large for any fixed  $n \geq 1$ ?
- Furthermore, given any constant  $C > 0$  and  $n \geq 1$ , are there always only finitely many totally real fields  $K$  such that  $U_{\mathcal{O}_K}(n+1) - U_{\mathcal{O}_K}(n) < C$ ? If so, can all such fields be effectively determined?
- For any fixed totally real field  $K$ , can we construct explicit criterion sets  $\mathcal{S}_K(n)$  for any  $\mathcal{O}_K$ -lattice to be  $n$ -universal?

**Thank you!**